

The Output Budget Confound: Why Token Ceilings Silently Decide LLM Planning Benchmark Results

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Abstract

Comparisons between large language models on planning benchmarks are usually reported as a single accuracy number per model. We show that for models which reason in the open, this number is not a property of the model alone: it is jointly determined by the output token ceiling the evaluator happened to choose. On the Calendar Scheduling task of *Natural Plan*, raising the ceiling from 512 to 4096 tokens moves Nemotron Super from **1.5%** to **84.6%**, a swing of 83 percentage points, with the prompt, the sample and the scoring held fixed. The model was not failing to plan. At the lower ceiling 98.5% of its generations were cut off before an answer was emitted, and an exact match scorer recorded that truncation as a planning error. We quantify the effect across tier matched models, separate it from test time compute scaling by reporting the truncation rate alongside accuracy, and show that the obvious way to detect truncation is unsound for models that reason in hidden tokens, failing in the direction of understating them. We release the harness and the per call records.

1 Introduction

Planning under constraints is an attractive target for language model evaluation: solutions are hard to find and cheap to verify. Natural Plan [1] scores a proposed meeting time by exact match against a golden answer, with no human annotation and no model acting as judge. That property is why we selected it. It removes the dominant failure mode of bespoke benchmarks, in which the party reporting a comparison also authored its ground truth.

Having removed that confound, we encountered another. Contemporary models differ sharply in how much text they emit before committing to an answer. A model that reasons silently answers in tens of tokens. A model that reasons in the open may spend thousands. Every evaluation harness imposes a generation ceiling. When that ceiling falls below what a model needs in order to finish, the harness records no answer, and an exact match scorer, which cannot distinguish an absent answer from an incorrect one, records a planning failure.

This is not hypothetical. In our first run, Nemotron Super scored zero of twelve at a ceiling of 512 tokens. Inspecting the raw generations showed the model mid derivation at the cut, having already correctly enumerated every participant constraint. At 4096 tokens, with nothing else changed, the same model on the same items scored 84.6%. Had we published the first number, we would have reported a false result in good faith, and the harness, not the model, would have been at fault.

1.1 Contributions

1. We isolate the generation ceiling as an experimental variable on Natural Plan Calendar Scheduling and measure its effect on two tier matched models chosen to sit at opposite ends of output verbosity.
2. We report the truncation rate and the solve rate conditional on finishing, which separate a planning failure from a harness failure and distinguish this effect from test time compute scaling [3].
3. We show that the obvious way to detect truncation, comparing emitted tokens against the ceiling, is unsound for

models reasoning in hidden tokens, and that it fails in the direction of understating them.

4. We validate our scorer against published numbers before reporting any comparison, and quantify run to run variance from a seeded replication.

2 Method

Task and scoring. We use the Calendar Scheduling split of Natural Plan: 1,000 items, each giving the availability of every participant and a required meeting duration, with exactly one valid slot. Difficulty is controlled by participant count, from two to seven. Scoring follows the official evaluator: a regular expression extracts the first Day, HH:MM - HH:MM span and compares day, start and end against the golden answer.

Scorer validation. Before comparing any models we validated our port of the official evaluator against the reference predictions shipped with the dataset. Our implementation scores those predictions at 48.9%, against 48% reported in the original work, so our numbers sit on the same scale as the published ones.

Models. The primary comparison is between two models matched by vendor tier and chosen because they sit at opposite ends of output verbosity, which is the axis the effect acts on: Nemotron Super, a sparse mixture of experts with 120B total and 12B active parameters, which reasons in its visible output; and Claude Haiku 4.5, which is comparatively terse. Only the first has a published parameter count, so we do not plot accuracy against parameter count, because doing so for the other would require inventing a number. A small set of Gemini 3.5 Flash results is reported as preliminary: its free tier permits twenty requests per day for that model, which is not a sample size.

Controls. Every model receives the byte identical user prompt taken from the dataset. One deliberate asymmetry exists and we state it plainly: Nemotron is sent the vendor documented reasoning control in a system turn, which the other models have no equivalent for. That control did not do what its name

suggests; the model still emitted extended derivations. Sampling temperature is zero wherever the vendor SDK exposes it. Latency is measured around the successful request only, excluding our own retry backoff, so rate limiting imposed by our billing tier is not charged to a model's speed. Items are sampled stratified by participant count, because the raw distribution is half two participant items. Uncertainty is reported as Wilson score intervals [5].

Infrastructure failures are not answers. Calls that never reached the model are excluded from every rate and reported separately. Under load the provider returned HTTP 429 and 503 at a substantial rate; a call that exhausts its retries produced no generation to score, and placing it in the denominator would record an infrastructure failure as a planning failure. Our own first analysis did exactly that before we caught it, which we take as evidence the mistake is easy rather than unusual.

3 Results

Figure 1(a) sweeps the generation ceiling with every other variable held fixed. Nemotron rises from 1.5% to 84.6%. Claude, which emits far less text before answering, is substantially flatter across the same sweep. A single ceiling applied uniformly is therefore not neutral: it penalises verbose reasoners specifically, and the size of the penalty is a property of the harness rather than of the task.

Figure 1(b) shows the mechanism. At the 512 token ceiling 98.5% of Nemotron generations terminate at the ceiling rather than at a stop token, against 46.2% for Claude. At 4096 that figure falls to 10.8%. The accuracy curve is the mirror image of the truncation curve, which is what distinguishes this from a test time compute effect: the model is not reasoning better with more tokens, it is being permitted to finish at all.

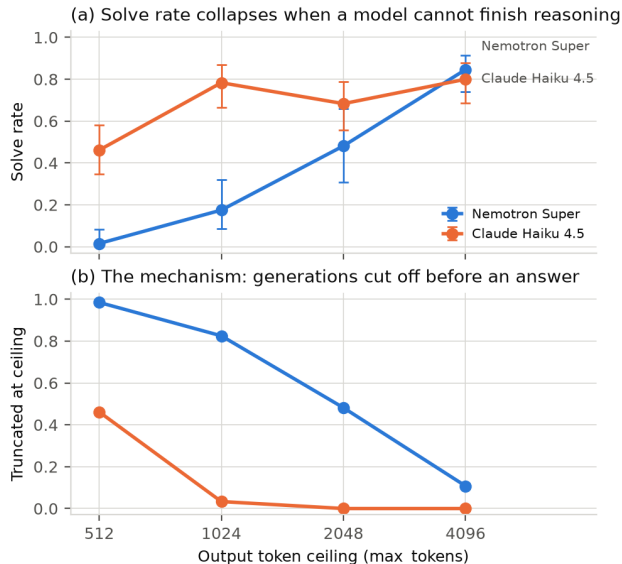


Figure 1. Solve rate against the output token ceiling (a), and the fraction of generations truncated at that ceiling (b). Bars are 95% Wilson intervals. The two panels are mirror images.

Model	Ceil	Solve	Trunc	Cond	n	Err
Nemotron	512	1.5	98.5	100.0	65	0
Nemotron	1024	17.5	82.5	85.7	40	20
Nemotron	2048	48.1	48.1	85.7	27	33
Nemotron	4096	84.6	10.8	94.8	65	0
Claude	512	46.2	46.2	82.9	65	0
Claude	1024	78.3	3.3	81.0	60	0
Claude	2048	68.3	0.0	68.3	60	0
Claude	4096	80.0	0.0	80.0	65	0

Table 1. Solve, truncation and conditional solve rates, in percent. Conditional is computed over generations allowed to finish. Err counts provider failures, excluded from every rate.

The conditional column is the one that matters for interpreting the first. A generation that is cut off and a generation proposing the wrong meeting time are the same event to an exact match scorer, but completely different events for anyone choosing a model. Separating them shows Nemotron's conditional accuracy never falls below 85.7% at any ceiling, while its headline number ranges across 83 points. The headline is measuring the harness.

We also resist the phrase *adequate ceiling* as an absolute. Nemotron remains truncated on 10.8% of items even at 4096 tokens, a ceiling most practitioners would consider generous, so its result there is still an underestimate. Adequacy is a property of a model and a harness together, and the only way to know a ceiling was adequate is to measure the truncation rate and find it at zero.

3.1 Detecting truncation is harder than it looks

The obvious way to detect that a generation hit the ceiling is to compare the emitted token count against it. That test is wrong for any model that reasons in hidden tokens. In our Gemini runs, generations were visibly cut in mid sentence, including mid answer, while the reported output token count stood at roughly twenty against a ceiling of 512.

Instrumenting the vendor's full token accounting shows why. At a ceiling where the model runs to completion it emits 21 visible tokens per item but spends 1344 reasoning tokens that do not appear in that count. True generation spend is roughly 65 times what the output token field reports.

```
ceiling 512
visible tokens    21 (what the API reports)
thinking tokens 1344 (billed, not reported)
true spend      ~1365 > 512, so truncation
                  is unavoidable
```

Truncation at the lower ceiling is therefore not merely possible, it is arithmetically unavoidable, which is exactly what the raw generations show. The reasoning tokens are billed against the ceiling but are absent from the field an evaluator would naturally check, so the naive test reports no truncation precisely where truncation is certain. We flag this because the failure is silent, it is in the direction that makes a model look worse, and it lands specifically on the reasoning models this effect

concerns. We take the vendor's finish reason as authoritative and treat any token based rule as a fallback.

3.2 What separates the models

At the highest ceiling Nemotron reaches 84.6% against Claude at 80.0%, and 94.8% against 80.0% conditional on finishing. Exact McNemar tests on the paired per item outcomes separate none of the model pairs. We report the comparison as not separable. Because all models are run on the same items the outcomes are paired, and a two sample proportion test would discard that pairing and overstate the evidence.

The efficiency picture is sharper than the accuracy one. Nemotron spends 2238 output tokens and 23.2 seconds per item against Claude's 547 tokens and 3.5 seconds, roughly four times the output and six times the wall clock for a solve rate whose interval overlaps. Accuracy alone does not separate these systems; accuracy per token does.

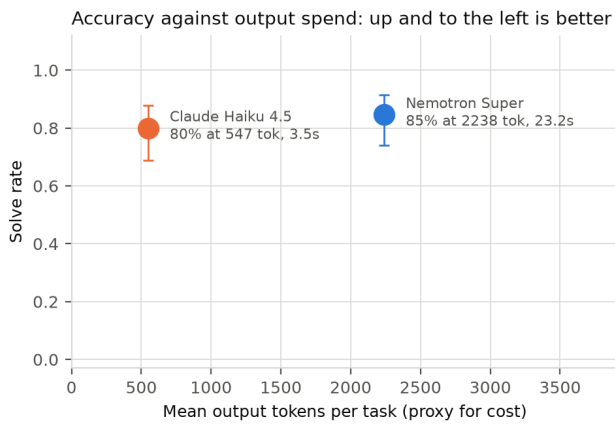


Figure 2. Solve rate against mean output tokens per task. Up and to the left is better. The models are close in accuracy and an order of magnitude apart in what they spend to get there.

Run to run variation reinforces the caution. An interrupted earlier sweep had already scored part of the seeded sample, so the relaunched sweep drew the same items and gave an independent replication at no extra cost. Nemotron, at temperature zero, scored 98.3% of items identically across the two runs. Claude, which we could not pin to temperature zero because its SDK no longer exposes the parameter, agreed on only 62.5%. A single accuracy figure for a model at a nondeterministic temperature is a draw from a distribution whose spread is not small.

4 Discussion

The practical recommendation is small and cheap. Evaluation reports should state the generation ceiling alongside accuracy, and should verify that the ceiling binds on no model before comparing any two. One statistic suffices: the fraction of generations that terminate at the ceiling rather than at a stop token. A model whose truncation rate is materially above zero has not been measured, it has been cut off, and its accuracy number carries no information about its planning ability.

The effect interacts badly with two current trends. Models increasingly reason at length, whether in visible output or in

hidden tokens, and harnesses increasingly cap output to control cost. Those pressures push in opposite directions, and the resulting headline number is quietly sensitive to a parameter that is rarely reported at all.

5 Limitations

Contamination is not controlled. Natural Plan has been public since 2024 and may appear in the training data of any model evaluated here. Exposure is symmetric across models and the headline result is a within model effect, but we make no decontamination claim.

Temperature. The Anthropic SDK version used removed the temperature parameter from its message creation call, so Claude runs at the API default while the others run at zero. This is an SDK constraint, stated rather than omitted.

Sample loss correlates with the swept variable. Provider 429 and 503 rates rose with the ceiling, because larger ceilings mean longer generations and more concurrent requests in flight. Higher ceiling cells therefore rest on fewer retained items. A dropped call produced no output at all, independent of whether that output would have been correct, so we do not believe this biases direction, but it costs precision where we would most like to have it. Re-running the highest ceiling at lower concurrency cut the loss rate substantially, confirming the cause.

Scope. One task from one benchmark, two models in the primary comparison. We do not claim the ceiling effect generalises to tasks that do not elicit extended reasoning, although the mechanism is not task specific.

6 Reproducibility

The harness, the per call records including raw generations, and the analysis code are released with this paper. Sampling is seeded and stratified, and the seed is recorded in every result file. Scoring uses our validated port of the official evaluator. Every number quoted here is generated from the records rather than typed by hand.

7 Conclusion

A generation ceiling is not a neutral implementation detail of an evaluation harness. For models that reason at length it can move a headline accuracy number by a wide margin, and it does so silently, because a truncated generation and a wrong answer are indistinguishable to an exact match scorer. We recommend that benchmark reports state the ceiling, verify it binds on no model, and publish output length alongside accuracy.

References

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